

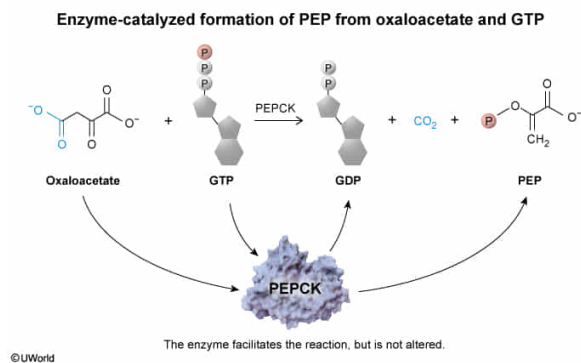
During gluconeogenesis, the enzyme phosphoenolpyruvate carboxykinase (PEPCK) catalyzes production of phosphoenolpyruvate (PEP) by decarboxylating oxaloacetate and transferring a phosphate group from GTP. What is the balanced equation for this reaction?

- ✔ ☒ A. Oxaloacetate + GTP → GDP + CO₂ + PEP [69%]
- ☐ B. Oxaloacetate + GTP + PEPCK → GDP + 2 CO₂ + PEP [12%]
- ☐ C. Oxaloacetate + GTP → GDP + 2 P_i + PEP [7%]
- ☐ D. Oxaloacetate + GTP + PEPCK → GDP + P_i + PEP [10%]

Correct

68% answered correctly

Explanation:



Enzymes catalyze biologically important chemical reactions **without being consumed**. In any chemical reaction, atoms are rearranged but not created or destroyed. Therefore, **all atoms must be accounted for** on both the **left and right sides** of the equation.

According to the question, phosphoenolpyruvate carboxykinase (PEPCK) catalyzes the formation of phosphoenolpyruvate (PEP) during **gluconeogenesis** through the following steps (note that reactants always appear on the left side of the equation whereas products appear on the right):

- During decarboxylation, CO₂ is removed from oxaloacetate (a reactant), yielding CO₂ as a product of the reaction.
- GTP (a reactant) transfers a phosphate group to the decarboxylated oxaloacetate, producing GDP and PEP.

Because PEPCK is an enzyme, it facilitates these steps but is not altered by the reaction. Enzymes may be written above or below a reaction arrow to show their participation, but they never appear on the reactant side (left) or product side (right) of the arrow.

Based on this information, the balanced equation is: **oxaloacetate + GTP → GDP + CO₂ + PEP**.

(Choice B) PEPCK is not altered by the reaction, and should not be shown in the equation. Additionally, the reaction produces *one* CO₂ molecule, not two.

(Choice C) The phosphate group from GTP is incorporated directly into PEP, so no free phosphate (P_i) is produced.

(Choice D) PEPCK should not appear in the reaction, and the reaction does not produce any inorganic phosphate (P_i).

Educational objective:

Balanced chemical reactions must account for all atoms involved on both sides of the equation. Enzymes facilitate chemical reactions without being altered by them, and do not appear on either side of the balanced equation.

Time spent:0 sec

Subject:
Biochemistry

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System:
Enzymes